Venkata Lakshmi Parimala Pasupuleti

AI/ML Engineer · Healthcare AI · Python · AWS · Azure · GCP · LangGraph · LLM Evaluation · Agentic Systems · EHR · FHIR · HIPAA · CI/CD · Production AI

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PROFESSIONAL SUMMARY

AI/ML engineer with 8 years of healthcare software production depth and 2 years hands-on agentic AI and LLM systems. At Entrata: **HUD compliance pipelines, eligibility and rent calculations, RabbitMQ async processing** for 100K+ housing units at 99.5% uptime — the kind of work where a wrong import affects thousands of residents' eligibility decisions. Built **Leasel**: 8-agent LangGraph compliance investigation platform with deterministic HITL, evaluated across 1,240 lease PDFs (93.6% precision, 87.9% F1). Built **AgentProbe**: open-source LLM production monitoring toolkit — 5-metric eval engine, pass@k harness, CI/CD regression gate. Built **AI on FHIR**: NLP platform integrating LLMs over FHIR EHR APIs with HIPAA-compliant RBAC. Currently building ML models and data pipelines at Blue Horse Digital. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present

AI/ML Engineer (Volunteer) · Virginia, USA

- Developing ML models and data analysis pipelines for software applications and business intelligence projects — data collection, feature engineering, model training, evaluation, and integration with application-facing APIs and reporting layers.

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer Intern — Agentic Systems · RAG · LangGraph · Compliance AI

- Built an agentic mission-planning system (LangGraph stateful graph: parse → plan → retrieve → validate → conditional routing) delivering a risk-aware compliance strategy: compliant missions route to export, non-compliant ones to rejection with cited regulation. Implemented RAG layer (FAA Part 107 corpus), function calling, structured outputs, validation guardrails. Complete product: ideation to deployed production system.

Entrata (PropTech / Real Estate SaaS)
Sep 2018 – Aug 2024

Senior Software Engineer — Healthcare-Grade Compliance Systems · ETL Pipelines · AWS · Python · Production AI

- **Owned HUD income-limit compliance pipeline for 7 years** — few thousand federal rows → 180,000 domain records powering eligibility, rent (HAP), and tax-credit calculations for 100K+ housing units at 99.5% uptime. Zero compliance violations. Every release included baseline validation, reconciliation, and rollback — wrong eligibility data means thousands of residents receive incorrect decisions.
- **Built RabbitMQ async processing system** for gross-rent updates — retry logic and idempotency guarantees eliminating double-processing failures; owned through production incidents and annual cycles. Led microservices redesign (Docker, K8s) — 50% scalability gain, 40% deployment reduction.
- **Led AWS S3 document migration with 100% PHI compliance** — Libsodium-encrypted pipelines, zero security incidents. Optimized 200+ SQL queries (30% gain). Mentored 5+ engineers, 300+ peer code reviews. Shipped 20+ modules across Python, PHP, PostgreSQL, JavaScript over 120+ Agile sprints.

George Mason University
Aug 2024 – Aug 2025

Research Associate & Teaching Assistant

- Built prediction models (R^2 = 0.977, 190+ countries); applied Isolation Forest for anomaly detection across 3 data pipelines; deployed ML inference API (Quart async Python, HuggingFace + PyTorch). TA for 600+ students: Python, OOP, SQL, data modeling.

Entrata India, Pune
Jan 2017 – Sep 2018

Associate / Trainee Software Engineer

- Built compliance modules (Python, Java, PostgreSQL); led AWS S3 migration resolving 90% of legacy space constraints; redesigned document storage architecture for hundreds of thousands of records.

PROJECTS

Leasel: Agentic AI Compliance Platform — HITL, Deterministic Audit, LLM Production Monitoring
HEALTHCARE-GRADE · LLM EVAL

Python · LangGraph (8-agent) · FastAPI · ChromaDB · OCR · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_rag

- 8-agent LangGraph StateGraph (Triage → Config → Historical → StructuredData → Reasoning → Validation → Human Escalation) with confidence-based HITL. **Core rule enforced in code**: LLM extracts and reasons; human or deterministic logic makes every regulated call — same principle required for clinical AI decisions. Evaluated via AgentProbe: **93.6% precision, 87.9% F1, pass@3 88.3%** across 1,240 lease PDFs. Complete product: ideation to production with CI regression gate.

AgentProbe: Open-Source LLM Production Monitoring & Eval Infrastructure
PRODUCTION AI · LLM MONITORING

Python · FastAPI · LangGraph · OpenAI · Anthropic · Groq · Ollama · CI/CD (GitHub Actions) — github.com/venkatalakshmiparimala/AgentProbe

- Built fully local, zero-cost LLM production monitoring toolkit: **5-metric eval engine** (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression harness with golden datasets, **model routing** across OpenAI/Anthropic/Groq/Ollama, offline + online evaluation loops. Artifact and dependency management via CI/CD pipeline — used to evaluate Leasel across 1,240+ compliance cases.

AI on FHIR: NLP Platform for EHR Medical Records
HEALTHCARE AI · FHIR · HIPAA

Flask · LLMs · FHIR API · BERT · Embeddings · React · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- ~60% reduction in manual clinical retrieval time — NLP platform translating natural-language queries (EN/ES) into **FHIR EHR API calls** using LLMs, BERT, and embeddings; enforced **HIPAA-compliant RBAC** with data isolation across structured medical records. Production-deployed, containerized with Docker.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

AI systems, ML modeling, data engineering, statistical modeling, cloud infrastructure

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017

TECHNICAL SKILLS

AI & Agents: LangChain, LangGraph, Agentic AI, Multi-Agent Orchestration, Tool Calling, Function Calling, Structured Outputs, RAG Pipelines, HITL, LLM Evaluation, LLM-as-Judge, Golden Datasets, Offline Evaluation, Online Evaluation, A/B Testing, Model Routing, Cost Controls, Guardrails, Hallucination Detection, Confidence Scoring, Auditability, Agent Observability, Trace-Level Observability, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama, ChromaDB, Qdrant, MCP, Production AI

Healthcare & Clinical AI: Healthcare AI, Clinical AI, EHR Integration, FHIR, HIPAA Compliance, Clinical NLP, Clinical Workflows, Patient Workflows, Ambient AI, Revenue Cycle Management (RCM), PHI Compliance, Regulatory Compliance

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure (ADF, ADLS, Databricks), GCP, Docker, Kubernetes, OpenShift, CI/CD (GitHub Actions), Terraform, Infrastructure-as-Code (IaC), Prometheus, Grafana, OpenTelemetry, RabbitMQ, Redis, Microservices, Distributed Systems

Languages & Backend: Python, Java, JavaScript, TypeScript, PHP, Scala, SQL, PL/SQL, R, Bash; FastAPI, Spring Boot, REST APIs; PostgreSQL, MySQL, Oracle, Snowflake, Teradata, NoSQL

ML & Data: PyTorch, TensorFlow, Scikit-learn, HuggingFace, XGBoost, Isolation Forest, Pandas, NumPy, spaCy, BERT, Tableau, Power BI, ETL, Data Pipelines, Apache Spark, PySpark